

$$3 \cdot 3 = 9 \tag{1}$$

$$a := 3 \tag{2}$$

$$3a = 9 \tag{3}$$

$$M := \begin{bmatrix} 3 & 1 & 5 \\ 4 & 2 & 6 \\ 5 & 3 & 7 \end{bmatrix} \tag{4}$$

$$3 \cdot M = \begin{bmatrix} 9 & 3 & 15 \\ 12 & 6 & 18 \\ 15 & 9 & 21 \end{bmatrix} \tag{5}$$

$$B := \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \tag{6}$$

$$M \cdot B = \begin{pmatrix} 29 \\ 38 \\ 47 \end{pmatrix} \tag{7}$$

$$f(x) := 5x^2 + 2x + x \tag{8}$$

$$f(\{5, 10\}) = \{140, 530\} \tag{9}$$

$$\{ \ x^2 \ = 9 \ = \{-3, 3\} \tag{10}$$

$$\left\{ \begin{array}{lcl} 2x + 5y + 2z & = & -38 \\ 3x - 2y + 4z & = & 17 \\ -6x + y - 7z & = & -12 \end{array} \right. = \begin{pmatrix} 3 \\ -8 \\ -2 \end{pmatrix} \tag{11}$$

$$3 = 3 \wedge 2 < 4 = 1 \tag{12}$$

$$relu(x) := \begin{cases} 0 & \text{if } x < 0 \\ x & \text{otherwise} \end{cases} \tag{13}$$

